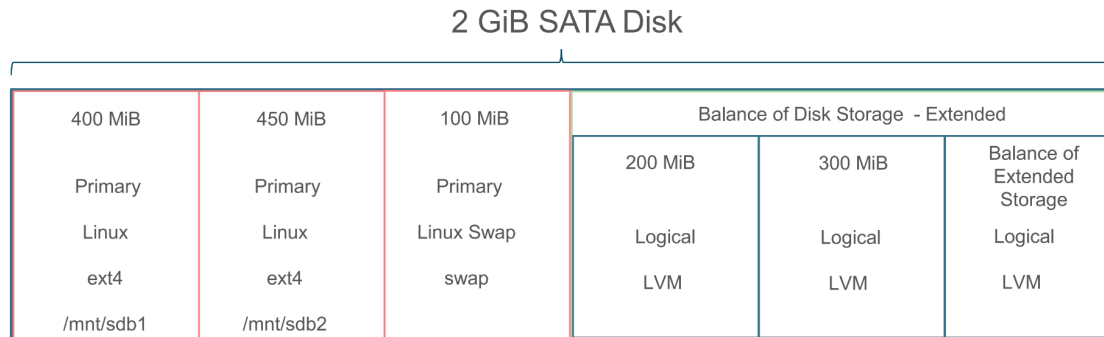
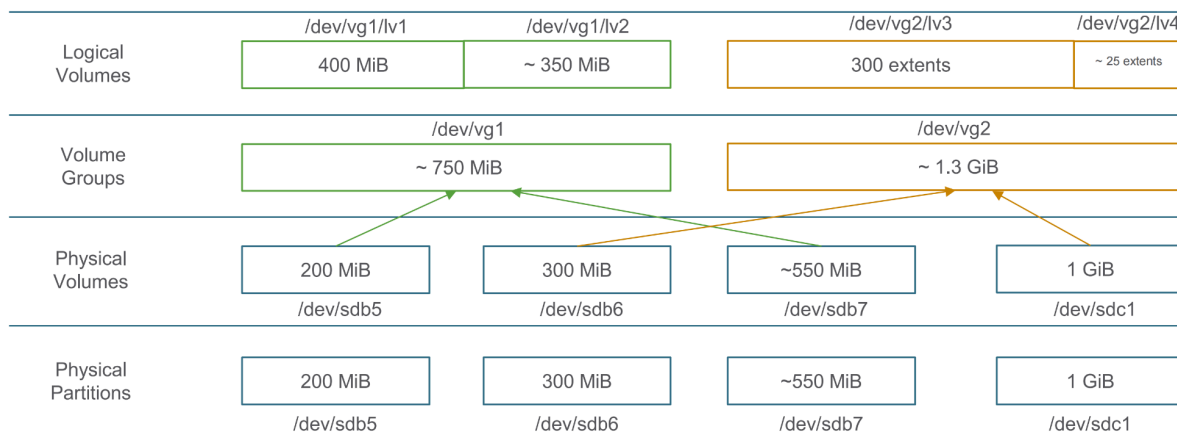


Practice Exercise (40 minutes) - Partitioning



Practice Exercise (30 minutes) - LVM



Week 12 practical 1 – Partitioning exercise

The final partitioning slide specifies a **2 GiB SATA disk** with this layout:

Partition	Type	Size	Use
<code>/dev/sdb1</code>	Primary Linux	400 MiB	ext4, mounted at <code>/mnt/sdb1</code>

<code>/dev/sdb2</code>	Primary Linux	450 MiB	ext4, mounted at <code>/mnt/sdb2</code>
<code>/dev/sdb3</code>	Primary Linux swap	100 MiB	Swap
<code>/dev/sdb4</code>	Extended	Remaining space	Container
<code>/dev/sdb5</code>	Logical Linux LVM	200 MiB	LVM
<code>/dev/sdb6</code>	Logical Linux LVM	300 MiB	LVM
<code>/dev/sdb7</code>	Logical Linux LVM	Remaining extended space	LVM

Step 1: Confirm the disk

```
lsblk
sudo fdisk -l /dev/sdb
```

Confirm that `/dev/sdb` is the empty 2 GiB disk.

Step 2: Create the partitions

```
sudo fdisk /dev/sdb
```

Inside `fdisk`, create `/dev/sdb1`:

```
n
p
1
Enter
+400M
```

Create `/dev/sdb2`:

```
n
p
2
```

Enter
+450M

Create `/dev/sdb3`:

n
p
3
Enter
+100M

Change `/dev/sdb3` to swap:

t
3
82

Create `/dev/sdb4` using the remaining space:

n
e
4
Enter
Enter

Create `/dev/sdb5`:

n
l
Enter
+200M

Create `/dev/sdb6`:

n
l
Enter
+300M

Create `/dev/sdb7` from the remaining extended space:

n
l
Enter
Enter

Set logical partitions to LVM:

```
t  
5  
8e  
t  
6  
8e  
t  
7  
8e
```

Review:

```
p
```

Write:

```
w
```

Reload and verify:

```
sudo partprobe /dev/sdb  
lsblk  
sudo fdisk -l /dev/sdb
```

Step 3: Format the Linux partitions

```
sudo mkfs.ext4 /dev/sdb1  
sudo mkfs.ext4 /dev/sdb2
```

Verify:

```
lsblk -f
```

Step 4: Configure swap

```
sudo mkswap /dev/sdb3  
sudo swapon /dev/sdb3
```

Verify:

```
swapon --show  
free -h
```

Step 5: Create mount points and mount

```
sudo mkdir -p /mnt/sdb1 /mnt/sdb2
sudo mount /dev/sdb1 /mnt/sdb1
sudo mount /dev/sdb2 /mnt/sdb2
```

Verify:

```
df -h
findmnt /mnt/sdb1
findmnt /mnt/sdb2
```

Step 6: Configure persistent mounting

Get UUIDs:

```
sudo blkid /dev/sdb1 /dev/sdb2 /dev/sdb3
```

Back up `/etc/fstab`:

```
sudo cp /etc/fstab /etc/fstab.backup
```

Add:

```
UUID=<sdb1-uuid> /mnt/sdb1 ext4 defaults 0 0
UUID=<sdb2-uuid> /mnt/sdb2 ext4 defaults 0 0
UUID=<sdb3-uuid> none swap defaults 0 0
```

Test:

```
sudo mount -a
sudo swapon -a
```

Verify:

```
findmnt /mnt/sdb1
findmnt /mnt/sdb2
swapon --show
```

Week 12 practical 2 – LVM exercise

The final LVM slide uses these physical partitions:

Physical partition	Approximate size
<code>/dev/sdb5</code>	200 MiB
<code>/dev/sdb6</code>	300 MiB
<code>/dev/sdb7</code>	550 MiB
<code>/dev/sdc1</code>	1 GiB

The required structure is:

```
vg1 ≈ 750 MiB
├── lv1 = 400 MiB
└── lv2 = remaining space, approximately 350 MiB
```

```
vg2 ≈ 1.3 GiB
├── lv3 = 300 extents
└── lv4 = remaining space, approximately 25 extents
```

Step 1: Prepare `/dev/sdc1`

Confirm the disk:

```
lsblk
sudo fdisk -l /dev/sdc
```

Create one partition using the full disk:

```
sudo fdisk /dev/sdc
```

Inside:

```
n
p
1
Enter
Enter
t
1
8e
p
w
```

Reload:

```
sudo partprobe /dev/sdc
```

Verify:

```
lsblk  
sudo fdisk -l /dev/sdc
```

Step 2: Create the physical volumes

```
sudo pvcreate /dev/sdb5 /dev/sdb6 /dev/sdb7 /dev/sdc1
```

Verify:

```
sudo pvs
```

Step 3: Create **vg1** and **vg2**

```
sudo vgcreate vg1 /dev/sdb5 /dev/sdb7  
sudo vgcreate vg2 /dev/sdb6 /dev/sdc1
```

Verify:

```
sudo vgs  
sudo vgdisplay vg1  
sudo vgdisplay vg2
```

Step 4: Create **lv1** and **lv2**

```
sudo lvcreate -L 400M -n lv1 vg1  
sudo lvcreate -l 100%FREE -n lv2 vg1
```

Verify:

```
sudo lvs
```

Step 5: Create **lv3** and **lv4**

```
sudo lvcreate -l 300 -n lv3 vg2  
sudo lvcreate -l 100%FREE -n lv4 vg2
```

Verify:

```
sudo lvs -o lv_name,vg_name,lv_size
```

Because LVM allocates by extents, the displayed sizes may not match the rounded diagram values exactly.

Step 6: Format and mount the logical volumes

```
sudo mkfs.ext4 /dev/vg1/lv1
sudo mkfs.ext4 /dev/vg1/lv2
sudo mkfs.ext4 /dev/vg2/lv3
sudo mkfs.ext4 /dev/vg2/lv4
```

Create mount points:

```
sudo mkdir -p /mnt/lv1 /mnt/lv2 /mnt/lv3 /mnt/lv4
```

Mount:

```
sudo mount /dev/vg1/lv1 /mnt/lv1
sudo mount /dev/vg1/lv2 /mnt/lv2
sudo mount /dev/vg2/lv3 /mnt/lv3
sudo mount /dev/vg2/lv4 /mnt/lv4
```

Verify:

```
lsblk -f
df -h
sudo pvs
sudo vgs
sudo lvs
```

Common Week 12 mistakes

Partitioning the wrong disk

Never copy this blindly:

```
sudo fdisk /dev/sda
```

Use:

```
lsblk
```


and identify the empty additional disk first.

Formatting the whole disk instead of the partition

Incorrect for this lab:

```
sudo mkfs.ext4 /dev/sdb
```

Correct:

```
sudo mkfs.ext4 /dev/sdb1
```

Forgetting to set partition types

The exercise requires:

- `/dev/sdb3` as swap
- `/dev/sdb5`, `/dev/sdb6`, `/dev/sdb7` and `/dev/sdc1` as LVM

Set these inside `fdisk` using `t`.

Formatting LVM partitions directly

Do not run:

```
sudo mkfs.ext4 /dev/sdb5
```

when `/dev/sdb5` is intended as an LVM physical volume.

The correct chain is:

`/dev/sdb5` → PV → VG → LV → filesystem

Format the LV, such as:

```
sudo mkfs.ext4 /dev/vg1/lv1
```

Forgetting to create the mount point

This fails:

```
sudo mount /dev/sdb1 /mnt/sdb1
```

when `/mnt/sdb1` does not exist.

Create it first:

```
sudo mkdir -p /mnt/sdb1
```

Using `mount` without configuring `/etc/fstab`

A manual mount normally disappears after reboot.

Add a correct persistent entry to:

`/etc/fstab`

Editing `/etc/fstab` and rebooting without testing

Always run:

```
sudo mount -a
```

before rebooting.

Confusing `-L` and `-l` in `lvcreate`

Uppercase `-L` specifies a size:

```
sudo lvcreate -L 400M -n lv1 vg1
```

Lowercase `-l` specifies extents or percentages:

```
sudo lvcreate -l 300 -n lv3 vg2
```

```
sudo lvcreate -l 100%FREE -n lv2 vg1
```

Extending an LV but not its filesystem

This enlarges only the logical volume:

```
sudo lvextend -L +200M /dev/vg1/lv1
```

For ext4, also run:

```
sudo resize2fs /dev/vg1/lv1
```

or combine both:

```
sudo lvextend -r -L +200M /dev/vg1/lv1
```

Week 12 essential commands

Command	Purpose
<code>lsblk</code>	Display disks, partitions and LVM devices
<code>lsblk -f</code>	Display filesystems and UUIDs
<code>fdisk -l</code>	List partition tables
<code>fdisk</code> <code>/dev/DISK</code>	Modify a disk partition table
<code>partprobe</code>	Ask the kernel to reread a partition table
<code>mkfs.ext4</code>	Create an ext4 filesystem
<code>blkid</code>	Display filesystem type and UUID
<code>mount</code>	Attach a filesystem
<code>umount</code>	Detach a filesystem
<code>findmnt</code>	Display mounted filesystems
<code>df -h</code>	Display filesystem usage
<code>mkswap</code>	Prepare swap space
<code>swapon</code>	Activate swap
<code>swapoff</code>	Deactivate swap
<code>free -h</code>	Display memory and swap usage
<code>pvccreate</code>	Create an LVM physical volume
<code>pvs</code>	Summarise physical volumes

<code>vgcreate</code>	Create a volume group
<code>vgextend</code>	Add storage to a volume group
<code>vgs</code>	Summarise volume groups
<code>lvcreate</code>	Create a logical volume
<code>lvextend</code>	Extend a logical volume
<code>lvs</code>	Summarise logical volumes
<code>resize2fs</code>	Expand an ext filesystem
<code>mount -a</code>	Test and mount <code>/etc/fstab</code> entries